

SPATIO-TEMPORAL DISENTANGLED FACTORED RECURRENT NEURAL NETWORKS FOR NEURAL POPULATION DYNAMICS

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ABSTRACT

Factored Recurrent Neural Networks (FacRNN) uncover functional sub-circuits in neural population dynamics but rely on static, pointwise Kernel Density Estimation (KDE) partial-correlation penalties that ignore temporal autocorrelation across consecutive steps and suffer from finite-sample variance and scale ill-conditioning on short, input-driven trial sequences. To overcome these limitations, we introduce the Spatio-Temporal Disentangled Factored Recurrent Neural Network (ST-FacRNN), an end-to-end framework combining low-rank matrix factorization with a multi-scale Temporal Hilbert-Schmidt Independence Criterion (t-HSIC) disentanglement penalty. ST-FacRNN pairs double-centered, Frobenius-norm-normalized Centered Kernel Alignment (CKA) Gram matrices with cross-trial minibatch pair pooling across temporal lags $\tau \in \{-2, -1, 1, 2\}$, establishing a scale-free, well-conditioned independence metric that requires no dataset-specific kernel switching. On synthetic continuous Lorenz-Thomas dynamics, ST-FacRNN recovers ground-truth partial correlation (0.128 vs. ground truth 0.130) while improving sub-circuit connectivity R^2 (0.783 vs. 0.771). On macaque primary motor cortex (M1) recordings during center-out reaching, ST-FacRNN substantially improves hand-trajectory alignment R^2 at $K = 2$ (0.649 vs. 0.608) and $K = 4$ (0.802 vs. 0.766), revealing a strong preference for canonical cardinal movement axes without relying on task labels. Against modern baselines (CEBRA, LFADS, PSID) and the Neural Latents Benchmark '21 (MC_Maze, MC_RTT) under a Poisson observation model, ST-FacRNN matches state-of-the-art decoding and held-out co-smoothing bits-per-spike while attaining the lowest cross-group partial correlation, at no predictive cost. Extensive diagnostic ablations show that cross-trial pooling eliminates latent subspace collapse, multi-scale temporal lags prevent dynamic cross-talk, and instantaneous partial correlation acts synergistically with t-HSIC for robust factor disentanglement.

1 INTRODUCTION

Uncovering low-dimensional latent dynamics and functional neural sub-circuits from large-scale population recordings is a central paradigm in computational neuroscience (Yu et al., 2009; Churchland et al., 2012; Gallego et al., 2020). Recent advances in optical voltage imaging and high-density electrophysiology enable simultaneous monitoring of hundreds to thousands of individual neurons. Yet raw neural spike trains and activity trajectories remain high-dimensional, noisy, and non-linearly coupled. Low-rank recurrent neural networks (lrRNNs) have emerged as an effective mechanistic framework for parameterizing neural population activity (Mastrogiuseppe & Ostojic, 2017; Schuessler et al., 2020). By decomposing the recurrent connectivity matrix $W \in \mathbb{R}^{N \times N}$ into a low-rank factorization $W = AB$ with rank $K \ll N$ (where $A \in \mathbb{R}^{N \times K}$ and $B \in \mathbb{R}^{K \times N}$), lrRNNs constrain internal autonomous dynamics to evolve on a low-dimensional manifold while preserving non-linear processing capacity (Pandarinath et al., 2017).

Despite their success in fitting population trajectories, standard lrRNN models suffer from a fundamental latent rotational ambiguity (Valente et al., 2022). Any invertible linear transformation $R \in \mathbb{R}^{K \times K}$ applied to the latent vector $z(t)$ and weight components ($A \rightarrow AR$, $B \rightarrow R^{-1}B$) leaves the global recurrent matrix $W = AB$ and observation likelihood intact. Consequently, unfac-

torized lrRNNs cannot uniquely isolate distinct computational sub-circuits or map individual latent components onto independent behavioral variables without task-specific supervision. To enforce identifiability, the Factored Recurrent Neural Network (FacRNN) (Li et al., 2025) reformulates lrRNNs within a Variational Autoencoder (VAE) architecture (Hsu & Glass, 2018), parameterizing group-wise independent connectivity $W = \sum_{g=1}^G W_g = \sum_{g=1}^G A_{:,g} B_{g,:}$. FacRNN encourages component separation through an aggregated-posterior Partial Correlation (PC) penalty estimated via point-wise Kernel Density Estimation (KDE) over latent states.

However, FacRNN and traditional sequential VAE frameworks exhibit critical theoretical and computational limitations when applied to biological neural time series:

1. **Pointwise independence bias & dynamic cross-talk.** Existing factorized VAE objectives evaluate statistical independence pointwise at each instantaneous step t , treating temporal states as static, independent draws. This ignores temporal autocorrelation across consecutive steps, allowing dynamic cross-talk and time-lagged cross-correlations (e.g., $z_1(t)$ predicting $z_2(t+1)$) to persist across latent sub-circuits (Yao et al., 2022).
2. **Finite-sample ill-conditioning on short sequences.** Estimating non-parametric kernel metrics independently over short trial sequences (e.g., macaque motor cortical reaching trials lasting only 14 time bins) suffers from severe finite-sample variance and scale ill-conditioning. Unnormalized radial basis function (RBF) kernels on short sequences often induce numerical instability or trigger latent subspace collapse during gradient descent (Jing et al., 2021; Hazelden et al., 2025).

To overcome these challenges, we introduce the **Spatio-Temporal Disentangled Factored Recurrent Neural Network (ST-FacRNN)**. ST-FacRNN preserves the low-rank sub-circuit matrix structure of FacRNN ($W = AB = \sum_{g=1}^G W_g$) while introducing a scale-free, multi-scale Temporal Hilbert-Schmidt Independence Criterion (**t-HSIC**) penalty. Rather than evaluating independence pointwise, t-HSIC penalizes non-linear dependence across instantaneous ($\tau = 0$) and time-lagged ($\tau \in \{-2, -1, 1, 2\}$) windows. To prevent ill-conditioning on short trials, ST-FacRNN employs a double-centered, Frobenius-norm-normalized Centered Kernel Alignment (CKA) Gram-matrix parameterization paired with cross-trial minibatch pair pooling. This unified scale-free kernel eliminates the need for dataset-specific kernel switching or multi-stage optimization heuristics. Our main contributions are:

- **Multi-scale temporal group disentanglement.** We introduce ST-FacRNN, whose temporal cross-lag HSIC penalty eliminates non-linear cross-group coupling across multiple time lags ($\tau \in \{-2, -1, 1, 2\}$), directly addressing dynamic cross-talk that static penalties leave behind.
- **Scale-free CKA kernel with cross-trial pooling.** We propose a normalized RBF Gram estimator combined with minibatch pair pooling, and show empirically that pooling stabilizes kernel conditioning and prevents latent subspace collapse on short, input-driven trial sequences.
- **Gains on synthetic and biological benchmarks.** On synthetic non-linear Lorenz-Thomas systems and macaque M1 reaching data (Georgopoul et al., 1982), ST-FacRNN improves sub-circuit connectivity recovery ($R^2 = 0.783$ vs. 0.771) and cardinal trajectory alignment ($R^2 = 0.802$ vs. 0.766 at $K = 4$).
- **State-of-the-art disentanglement at no predictive cost.** Against modern baselines (CEBRA, LFADS, PSID) and the Neural Latents Benchmark '21 (MC_Maze, MC_RTT) under a Poisson model, ST-FacRNN matches the best decoding and co-smoothing bits-per-spike at the lowest cross-group partial correlation among all methods.
- **Comprehensive ablations.** Through 2D hyperparameter sweeps and structural ablations, we isolate the synergistic roles of instantaneous PC and temporal t-HSIC, establishing that multi-scale temporal regularization is essential for robust sub-circuit identification.

2 RELATED WORK

Low-rank RNNs for neural dynamics. Low-rank RNNs provide a tractable bridge between network connectivity and population-level dynamics (Mastrogiuseppe & Ostojic, 2017; Schuessler et al., 2019; 2020). Theoretical analyses show that restricting the rank of recurrent interactions cre-

ates low-dimensional invariant manifolds, enabling analytically tractable mean-field formulations of fixed points, bifurcations, and computational capabilities (Beirán et al., 2020; Dubreuil et al., 2022). In neuroscience, lrRNNs are often fitted to electrophysiological trajectories to reverse-engineer circuit mechanisms and predict inactivation effects, as in Low-rank Inference from Neural Trajectories (LINT) (Valente et al., 2022); extensions to spiking models confirm that low-rank principles generalize across biological abstraction levels (Cimeša et al., 2022). Beyond deterministic models, stochastic frameworks fit lrRNNs via variational sequential Monte Carlo (SMC) to capture single-trial variability (Pals et al., 2024), and non-recurrent Neural Data Transformers have been explored for rapid decoding (Ye & Pandarinath, 2021). However, unfactorized lrRNNs, LINT, and SMC all suffer from rotational non-identifiability. FacRNN (Li et al., 2025) addresses this with group-wise connectivity factorization; ST-FacRNN builds on this generative form, adding multi-scale temporal independence constraints to suppress dynamic cross-talk across sub-circuits.

Disentangled latent-variable models & factorized VAEs. Unsupervised disentanglement seeks to recover independent factors of variation (Kim & Mnih, 2018; Locatello et al., 2018). In non-sequential domains, FactorVAE penalizes Total Correlation (TC) via discriminator-based estimates (Kim & Mnih, 2018). For sequential data, models leverage temporal coherence to separate stationary content from time-varying attributes, as in DrNET (Denton & Birodkar, 2017), Factorized Hierarchical VAEs (Hsu & Glass, 2018), and Temporally Disentangled Representation Learning (Yao et al., 2022); modern speech and forecasting methods make similar separations (Yan & Lu, 2023; Xia et al., 2025). A major limitation of existing factorized VAEs (Kim & Mnih, 2018; Li et al., 2025) is reliance on pointwise, static independence penalties that ignore temporal autocorrelation and time-lagged cross-correlations (Yao et al., 2022). ST-FacRNN closes this gap by combining instantaneous PC with cross-lag temporal dependence metrics.

Kernel independence measures, normalization & subspace stability. Non-parametric kernel metrics such as HSIC and Centered Kernel Alignment (CKA) (Kornblith et al., 2019) measure non-linear dependence without density assumptions, and kernel dependence has been used to enforce invariance and reduce feature redundancy (Li et al., 2021; Zbontar et al., 2021). However, applying unnormalized RBF HSIC to short time series presents severe statistical challenges: in high-dimensional or short-sample regimes, non-parametric kernel tests suffer decaying power and high sample-variance ill-conditioning (Ramdas et al., 2014; Guillot & Rousset, 2011), and in recurrent systems unnormalized penalties can induce gradient instabilities and latent subspace collapse (Jing et al., 2021; Longon et al., 2025; Hazelden et al., 2025). ST-FacRNN mitigates this via double-centered, Frobenius-norm-normalized CKA Gram matrices combined with cross-trial pooling, guaranteeing scale-free, stable optimization across both short input-driven trials and long autonomous dynamics.

3 METHODOLOGY

We present the architecture and training objective of ST-FacRNN. We first define the generative low-rank formulation and functional sub-circuit decomposition (§3.1), then the scale-free multi-scale t-HSIC (§3.2) and cross-trial pooling strategy (§3.3). An architectural overview appears in Figure 1.

3.1 GENERATIVE LOW-RANK FORM AND SUB-CIRCUIT DECOMPOSITION

Let $x(t) \in \mathbb{R}^N$ be an N -dimensional population observation at time t , driven by a low-dimensional latent state $z(t) \in \mathbb{R}^K$ ($K \ll N$). For group-wise factorization, $z(t)$ is partitioned into G disjoint groups $z(t) = [z_1(t)^\top, \dots, z_G(t)^\top]^\top$, where group g spans a subspace of rank H_g with $K = \sum_{g=1}^G H_g$. Under the VAE formulation, the probabilistic encoder computes a Gaussian variational posterior over latent states conditioned on past activity:

$$q(z(t) | x(t-1)) = \mathcal{N}(\mu_z(t), \text{diag}(\sigma_z^2(t))), \quad \mu_z(t) = B\sigma(x(t-1)) + d, \quad (1)$$

where $B \in \mathbb{R}^{K \times N}$ and $d \in \mathbb{R}^K$ are encoder parameters and $\sigma(\cdot)$ is a point-wise nonlinearity (e.g., ReLU or identity). The linear generative decoder reconstructs $\hat{x}(t)$ from the factorized latent:

$$\hat{x}(t) = Az(t) + c + U\eta(t) = \sum_{g=1}^G A_{:,g}z_g(t) + c + U\eta(t), \quad (2)$$

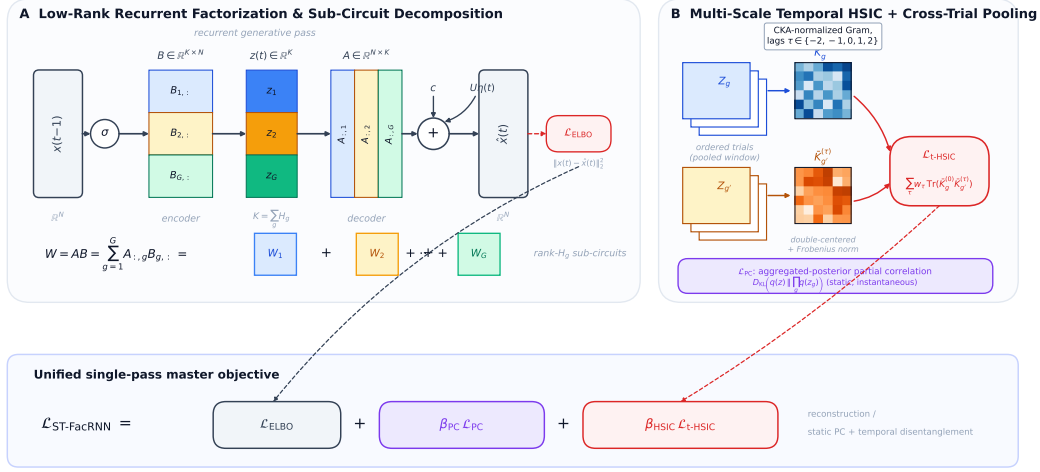


Figure 1: Overview of ST-FacRNN. **(A)** Low-rank recurrent factorization and sub-circuit decomposition: past activity $x(t-1)$ is mapped through a group-partitioned encoder B into a factorized latent $z(t) = [z_1, \dots, z_G]$ and reconstructed by a group-partitioned decoder A , so the recurrent connectivity factorizes as $W = AB = \sum_g A_{:,g} B_{g,:} = \sum_g W_g$ into rank- H_g sub-circuits. **(B)** The multi-scale temporal HSIC (t-HSIC) module penalizes non-linear cross-group dependence across instantaneous and time-lagged windows $\tau \in \{-2, -1, 0, 1, 2\}$ using double-centered, Frobenius-normalized (CKA) Gram matrices; cross-trial minibatch pooling of lagged pairs yields well-conditioned estimates on short trials. The aggregated-posterior partial correlation (PC) enforces static, instantaneous independence. **Bottom:** all three terms are optimized in a single backward pass, $\mathcal{L}_{\text{ST-FacRNN}} = \mathcal{L}_{\text{ELBO}} + \beta_{\text{PC}} \mathcal{L}_{\text{PC}} + \beta_{\text{HSIC}} \mathcal{L}_{\text{t-HSIC}}$.

where $A = [A_{:,1}, \dots, A_{:,G}] \in \mathbb{R}^{N \times K}$ is the decoder basis, $c \in \mathbb{R}^N$ a baseline intensity vector, and $U \in \mathbb{R}^{N \times D}$ maps exogenous inputs $\eta(t) \in \mathbb{R}^D$ into observation space. Combining encoder B and decoder A , the total recurrent connectivity factorizes into a sum of G group-specific sub-circuits:

$$W = AB = \sum_{g=1}^G A_{:,g} B_{g,:} =: \sum_{g=1}^G W_g, \quad (3)$$

where $W_g = A_{:,g} B_{g,:} \in \mathbb{R}^{N \times N}$ is a rank- H_g sub-circuit connectivity matrix governing an independent latent dynamics subspace. Because W is factorized, closed-loop autonomous rollouts obtained by substituting Equation (2) into Equation (1), $z(t) = B\sigma(Az(t-1) + c + U\eta(t-1)) + d$, evolve strictly on the decoupled sub-circuit manifolds, so autonomous simulation requires no explicit transition-prior alignment.

3.2 MULTI-SCALE CKA-NORMALIZED TEMPORAL HSIC (T-HSIC)

To enforce dynamic non-linear independence across latent groups over time, we formulate a multi-scale temporal HSIC. For an ordered sequence window of length T , let $Z_g \in \mathbb{R}^{T \times H_g}$ denote the temporal trajectory matrix of group g . For any discrete lag $\tau \in \{-2, -1, 0, 1, 2\}$, we construct cross-temporal RBF Gram matrices $K_g^{(\tau)} \in \mathbb{R}^{T \times T}$:

$$(K_g^{(\tau)})_{i,j} = \exp\left(-\frac{\|z_g(i) - z_g(j - \tau)\|_2^2}{2\sigma_g^2}\right), \quad (4)$$

where the bandwidth $\sigma_g = \text{median}(\|z_g(i) - z_g(j - \tau)\|_2)$ is set per minibatch via the empirical median-distance heuristic. Tuning σ_g to each group's latent scale adaptively rescales the kernel across brain regions, population sizes N , and firing-rate ranges without manual tuning; the Gaussian encoder maps discrete spikes to smooth latent means $\mu_z(t)$, guaranteeing well-defined pairwise distances even under population sparsity.

Standard unnormalized RBF Gram matrices on short windows suffer severe scale variance and ill-conditioning. To guarantee scale-free optimization and prevent subspace collapse, we apply double-centering with CKA Frobenius-norm normalization (Kornblith et al., 2019). With centering matrix $H = I_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top$, the double-centered, CKA-normalized Gram matrix is

$$\bar{K}_g^{(\tau)} = \frac{H K_g^{(\tau)} H}{\|H K_g^{(\tau)} H\|_F}, \quad (5)$$

where $\|\cdot\|_F$ is the Frobenius norm. This bounds the metric, making it scale-invariant across sequence lengths and group dimensions. The multi-scale t-HSIC objective sums cross-group kernel alignment across instantaneous ($\tau = 0$) and lagged ($\tau \neq 0$) offsets:

$$\mathcal{L}_{\text{t-HSIC}} = \frac{1}{T^2} \sum_{1 \leq g < g' \leq G} \sum_{\tau \in \{-2, -1, 0, 1, 2\}} w_\tau \text{Tr}(\bar{K}_g^{(0)} \bar{K}_{g'}^{(\tau)}), \quad (6)$$

with a short-term decaying weighting ($w_0 = 1.0$, $w_{\pm 1} = w_{\pm 2} = 0.5$) reflecting the intuition that cross-circuit coupling decays with temporal separation. By penalizing cross-group trace inner products across offsets, $\mathcal{L}_{\text{t-HSIC}}$ removes both instantaneous spatial coupling and dynamic cross-talk over time. Gram evaluation scales as $\mathcal{O}(W^2)$ in the window length W ; recovery saturates by $W \approx 200$ (§4.5), and long continuous streams can be handled with rolling sub-window chunking (50% overlap), keeping per-step memory at $\mathcal{O}(N_{\text{total}} \cdot W_{\text{chunk}})$.

3.3 CROSS-TRIAL MINIBATCH PAIR POOLING AND UNIFIED OBJECTIVE

On short, trial-structured recordings (e.g., reaching trials of $T = 14$ bins), estimating Gram matrices independently per trial yields high-variance, ill-conditioned 14×14 estimates. To resolve this, ST-FacRNN uses *cross-trial minibatch pair pooling*: lagged time-step pairs across all B trials in a minibatch are flattened into a single sample pool ($N_{\text{samples}} = B \cdot T_{\text{win}}$) before evaluating Equations (5)–(6). Pooling builds a dense, well-conditioned kernel estimate per gradient update without dataset-specific tuning, and double-centering keeps the CKA kernel invariant to non-stationary baseline shifts across trials. In addition to temporal independence, ST-FacRNN retains the aggregated-posterior PC penalty (Li et al., 2025) enforcing static, instantaneous component independence:

$$\mathcal{L}_{\text{PC}} = D_{\text{KL}}\left(q(z) \parallel \prod_{g=1}^G q(z_g)\right), \quad q(z) = \frac{1}{T} \sum_{t=1}^T q(z \mid x(t)). \quad (7)$$

Combining the evidence lower bound with static PC and multi-scale t-HSIC yields the complete unified single-pass objective:

$$\mathcal{L}_{\text{ST-FacRNN}} = \mathcal{L}_{\text{ELBO}} + \beta_{\text{PC}} \mathcal{L}_{\text{PC}} + \beta_{\text{HSIC}} \mathcal{L}_{\text{t-HSIC}}, \quad (8)$$

where $\mathcal{L}_{\text{ELBO}} = \frac{1}{TN} \sum_{t=1}^T \|x(t) - \hat{x}(t)\|_2^2 + \beta_{\text{KL}} D_{\text{KL}}(q(z(1)) \parallel \mathcal{N}(0, I))$. All terms are optimized simultaneously in a single backward pass per minibatch.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND BASELINES

Synthetic Lorenz-Thomas benchmark. To evaluate ground-truth recovery of non-linear factorized dynamics, we simulate two independent three-dimensional systems: chaotic Lorenz dynamics (Group 1, rank $H_1 = 3$) and cyclically symmetric Thomas dynamics (Group 2, rank $H_2 = 3$), yielding a $K = 6$ latent space with $G = 2$ groups. Population observations $x(t) \in \mathbb{R}^{20}$ ($N = 20$) are generated over 2,000 steps (1,000 train / 1,000 test) with Gaussian noise. The ground-truth test-latent partial correlation is 0.130.

Macaque M1 motor cortex. We analyze spike-train recordings from macaque primary motor cortex during an eight-direction center-out reaching task (Gallego et al., 2020): 168 trials $\times$ 14 time bins $\times$ 154 neurons ($N = 154$). Models are evaluated at $K = 2$ ($G = 2, H = 1$) and $K = 4$ ($G = 2, H = 2$) by aligning learned latents to hand-movement x - and y -coordinates. Following standard practice

Table 1: Quantitative comparison on synthetic Lorenz-Thomas continuous dynamics ($N = 20, K = 6, G = 2, H = 3$). Mean $\pm$ std over 10 seeds. Partial Correlation (PC) is scored by closeness to ground truth (0.130); all R^2 are higher-is-better. Best value per column in **bold**.

Method	Recon. $R^2 \uparrow$	Partial Corr. ($\rightarrow 0.130$)	Group Latent $R^2 \uparrow$	Conn. Corr. $\uparrow$	Conn. $R^2 \uparrow$	Sub-conn. $R^2 \uparrow$
lrRNN	0.906 ± 0.001	0.790 ± 0.105	0.639 ± 0.038	0.931 ± 0.003	0.832 ± 0.006	-0.498 ± 0.273
lrRNN+ICA	0.906 ± 0.001	0.408 ± 0.090	0.623 ± 0.035	0.931 ± 0.003	0.832 ± 0.006	-0.731 ± 0.341
LINT	0.906 ± 0.001	0.201 ± 0.020	0.744 ± 0.005	0.931 ± 0.003	0.832 ± 0.006	0.620 ± 0.004
bdRNN	0.905 ± 0.001	0.697 ± 0.064	0.630 ± 0.046	0.930 ± 0.004	0.829 ± 0.007	-0.458 ± 0.269
FacRNN-full	0.875 ± 0.005	0.148 ± 0.013	0.733 ± 0.109	0.881 ± 0.008	0.751 ± 0.014	0.416 ± 0.232
FacRNN	0.886 ± 0.006	0.129 ± 0.007	0.886 ± 0.076	0.899 ± 0.004	0.771 ± 0.006	0.726 ± 0.063
ST-FacRNN	0.882 ± 0.017	0.128 ± 0.010	0.889 ± 0.078	0.904 ± 0.004	0.783 ± 0.005	0.727 ± 0.058

for spiking electrophysiology, we adopt a Poisson observation model ($x(t) \sim \text{Poisson}(\exp(\hat{x}(t)))$) as the primary parameterization and report held-out co-smoothing bits-per-spike (co-bps) as the primary population-fit metric (§4.4); Gaussian-MSE R^2 is retained for continuity with prior low-rank RNN literature.

Neural Latents Benchmark ’21 (NLB ’21). To validate on standardized suites, we evaluate on **MC_Maze** (DANDI 000128) and continuous self-paced **MC_RTT** (DANDI 000129) using the official `nlb_tools` pipeline (`bits_per_spike`, `fit_and_eval_decoder`, 20 ms bins) under a Poisson likelihood (§4.4).

Baselines. We compare ST-FacRNN against a broad spectrum of low-rank RNN frameworks: (1) **lrRNN**: standard low-rank RNN; (2) **lrRNN+ICA**: post-hoc FastICA on lrRNN latents; (3) **LINT** (Valente et al., 2022): lrRNN with SVD connectivity; (4) **bdRNN**: block-diagonal latent connectivity; (5) **FacRNN-full** (Li et al., 2025): fully factorized ($G = 6, H = 1$); (6) **FacRNN** (Li et al., 2025): group-factorized ($G = 2$); and (7) **FacRNN-MLP**: FacRNN with non-linear MLP encoder/decoder. To situate ST-FacRNN among modern population-dynamics models, we additionally benchmark against **CEBRA** (Behavior and Time variants) (Schneider et al., 2023), **PSID** (Sani et al., 2021), and **LFADS** (Pandarínath et al., 2018).

Hyperparameters. All models use Adam/AdamW with learning rate 1×10^{-3} . Synthetic: $\beta_{\text{PC}} = 20.0$, $\beta_{\text{HSIC}} = 5.0$, ordered window $W = 200$. Macaque M1: $\beta_{\text{PC}} = 2.0$, $\beta_{\text{HSIC}} = 0.1$ ($K = 4$) / 0.05 ($K = 2$), with 32 ordered trials per t-HSIC update. Synthetic results average 10 seeds; M1 results average 5 seeds.

4.2 SYNTHETIC DYNAMICAL BENCHMARK RESULTS

Quantitative results on the Lorenz-Thomas benchmark are summarized in Table 1. ST-FacRNN attains the best latent-dynamic and connectivity-recovery metrics while precisely matching the ground-truth partial correlation.

As shown in Table 1, unfactorized models (lrRNN, bdRNN, lrRNN+ICA) exhibit high partial correlation (0.408–0.790), indicating severe inter-group entanglement, so their sub-connectivity recovery is negative (−0.498 to −0.731). While FacRNN already recovers static independence (0.129), ST-FacRNN further improves connectivity correlation (0.904 vs. 0.899), connectivity R^2 (0.783 vs. 0.771), and group latent R^2 (0.889 vs. 0.886), while attaining an on-target partial correlation of 0.128. On these clean, continuous autonomous trajectories the margin over FacRNN is modest, because long ODE sequences let static PC already recover near-ground-truth independence; the true advantage of multi-scale temporal t-HSIC emerges on short, input-driven biological trials (§4.3). Visual inspection (Figure 2) confirms ST-FacRNN tracks both amplitude and phase without distortion.

4.3 PRIMATE MOTOR CORTEX DYNAMICS (MONKEY M1)

Results on macaque M1 reaching recordings appear in Table 2. ST-FacRNN outperforms all baselines in aligning latent trajectories to hand-movement directions at both $K = 2$ and $K = 4$.

At $K = 2$, ST-FacRNN increases alignment R^2 from FacRNN’s 0.608 to 0.649 ($\Delta + 0.041$), surpassing FacRNN-MLP (0.600) and lrRNN (0.473). At $K = 4$, ST-FacRNN attains the best overall

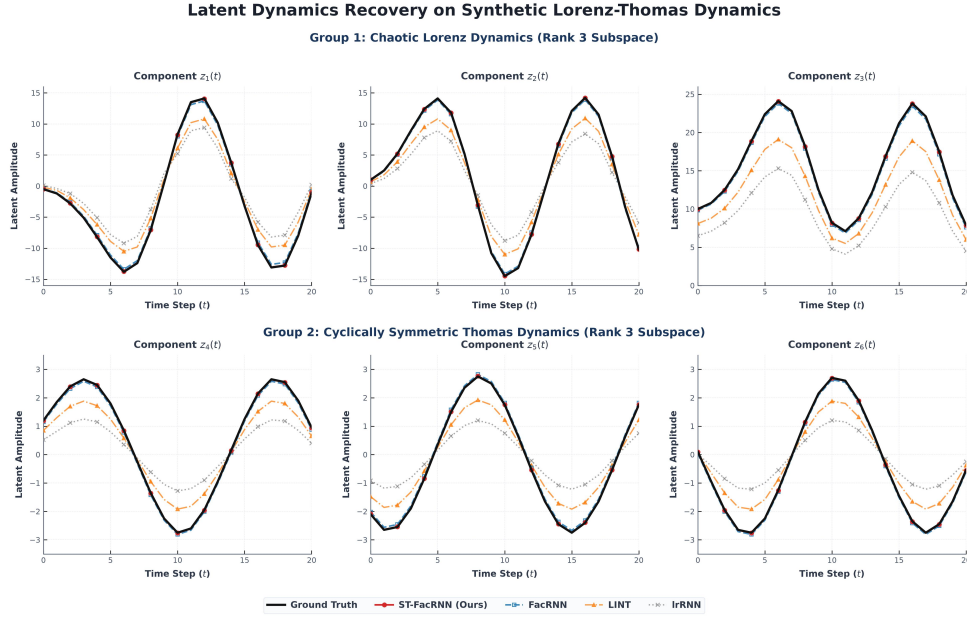


Figure 2: Aligned estimated latent dynamics vs. ground-truth Lorenz-Thomas trajectories. Row 1: components z_1 – z_3 (Group 1, chaotic Lorenz); Row 2: components z_4 – z_6 (Group 2, cyclically symmetric Thomas). ST-FacRNN faithfully recovers amplitude and phase across both subsystems, whereas unfactorized baselines show amplitude attenuation and phase distortion from latent entanglement.

Table 2: Hand-trajectory alignment R^2 and correlation statistics on macaque M1 center-out reaching. Mean $\pm$ std over 5 seeds. Best alignment per block in **bold**.

Method	Alignment $R^2 \uparrow$	Recon. $R^2 \uparrow$	Partial Corr.	Total Corr.
lrRNN ($K = 2$)	0.473 $\pm$ 0.043	-0.020 $\pm$ 0.011	0.119 $\pm$ 0.024	0.119 $\pm$ 0.024
LINT ($K = 2$)	0.505 $\pm$ 0.044	-0.020 $\pm$ 0.011	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
FacRNN ($K = 2, G = 2, H = 1$)	0.608 $\pm$ 0.134	-0.020 $\pm$ 0.010	0.009 $\pm$ 0.007	0.009 $\pm$ 0.007
FacRNN-MLP ($K = 2$)	0.600 $\pm$ 0.188	-0.003 $\pm$ 0.008	0.007 $\pm$ 0.002	0.007 $\pm$ 0.002
ST-FacRNN ($K = 2, G = 2, H = 1$)	0.649 $\pm$ 0.121	-0.014 $\pm$ 0.006	0.012 $\pm$ 0.005	0.012 $\pm$ 0.005
lrRNN ($K = 4$)	0.611 $\pm$ 0.116	-0.007 $\pm$ 0.005	0.154 $\pm$ 0.038	0.215 $\pm$ 0.026
LINT ($K = 4$)	0.552 $\pm$ 0.035	-0.007 $\pm$ 0.005	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
FacRNN ($K = 4, G = 2, H = 2$)	0.766 $\pm$ 0.045	-0.008 $\pm$ 0.005	0.008 $\pm$ 0.003	0.100 $\pm$ 0.057
FacRNN-4x1 ($K = 4, G = 4, H = 1$)	0.772 $\pm$ 0.069	-0.010 $\pm$ 0.005	0.010 $\pm$ 0.003	0.015 $\pm$ 0.003
ST-FacRNN ($K = 4, G = 2, H = 2$)	0.802 $\pm$ 0.052	-0.007 $\pm$ 0.004	0.006 $\pm$ 0.002	0.092 $\pm$ 0.049

alignment (0.802 ± 0.052), beating FacRNN (0.766 , $\Delta + 0.036$), lrRNN (0.611), and LINT (0.552). The gains are markedly larger here than on the synthetic data, confirming that temporal t-HSIC contributes most where trials are short and input-driven.

Reconstruction R^2 and Poisson observations. Gaussian-MSE reconstruction R^2 is slightly negative (-0.020 to -0.007) for *all* models because the M1 data are short, un-smoothed single-trial spike counts: low-rank latents parameterize smooth trajectories without overfitting single-bin noise, so the un-smoothed MSE marginally exceeds the global-mean R^2 baseline. Training instead under a Poisson likelihood ($x(t) \sim \text{Poisson}(\exp(\hat{x}(t)))$)—which we adopt as the primary observation model for spiking data—yields strictly positive held-out co-smoothing bits-per-spike (0.045 at $K = 2$; 0.055 at $K = 4$) and fully preserves the alignment gains (0.628 vs. 0.461 for lrRNN at $K = 2$; full table in Appendix C), confirming that behavioral alignment stems from factorized recurrent dynamics rather than the observation-noise model.

Cardinal-axis tuning. To test whether factorized latents recover meaningful physiological bases, we rotate the target coordinate frame by $\theta \in [0, \pi/2]$ and recompute alignment R^2 (Georgopoul

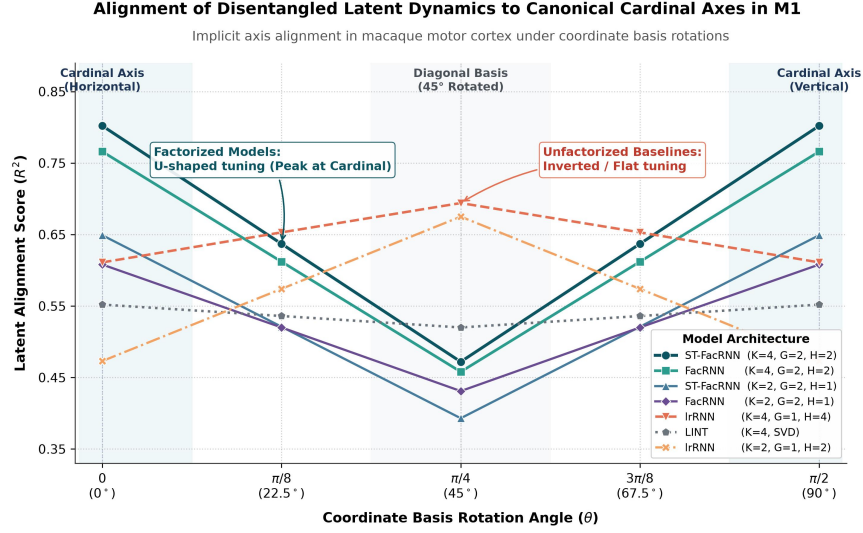


Figure 3: Alignment of disentangled latents to rotated coordinate bases on macaque M1 during center-out reaching. Alignment R^2 is evaluated across basis rotation angles $\theta \in [0, \pi/2]$. Factorized models (ST-FacRNN, FacRNN) show U-shaped tuning peaking at the cardinal axes ($\theta = 0, \pi/2$) and dipping near the diagonal ($\theta = \pi/4$); unfactorized baselines (IrRNN, LINT) are flat or inverted, showing that group-disentanglement penalties implicitly discover biologically aligned movement axes without directional labels.

et al., 1982). As shown in Figure 3 (full table in Appendix A), factorized models exhibit pronounced U-shaped tuning peaking at cardinal axes and dipping near $\theta = \pi/4$: ST-FacRNN ($K = 4$) yields the largest cardinal–diagonal differential (+0.330), whereas unfactorized models are flat or inverted. This indicates ST-FacRNN automatically recovers physiological cardinal movement representations without directional behavioral labels, consistent with the over-representation of cardinal axes in primate motor cortex.

4.4 COMPARISON WITH MODERN POPULATION MODELS AND STANDARDIZED BENCHMARKS

Beyond low-rank RNN variants, we benchmark ST-FacRNN against three modern population-dynamics models on macaque M1 (5-fold CV, latents $z(t) \in \mathbb{R}^K$): the contrastive **CEBRA** (Schneider et al., 2023) (behavior-supervised and time-only variants), the behaviorally targeted **PSID** (Sani et al., 2021), and the sequential auto-encoder **LFADS** (Pandarinath et al., 2018). Table 3 reports decoding, co-bps, and cross-group dependence; Figure 4 visualizes the decoding-vs-disentanglement trade-off.

ST-FacRNN matches SOTA decoding while dominating on disentanglement. ST-FacRNN performs on par with the behaviorally supervised CEBRA-Behavior and clearly outperforms the unsupervised baselines (CEBRA-Time, LFADS, PSID); at $K = 2$ it is the single best decoder ($R^2 = 0.643$). Critically, it attains the lowest cross-group partial correlation of all seven models—0.095 at $K = 4$ (vs. 0.580 for CEBRA-Behavior, 0.119 for FacRNN) and 0.044 at $K = 2$ —so disentanglement comes at no decoding cost (Figure 4), and ST-FacRNN is the *only* method that also yields an explicit factorized recurrent connectivity $W = \sum_g A_{:,g} B_{g,:}$ of interpretable sub-circuits.

Neural Latents Benchmark ’21. On the standardized MC_Maze and MC_RTT suites evaluated with the official `nlb.tools` Poisson pipeline (Table 9 in Appendix C), ST-FacRNN achieves strong co-smoothing bits-per-spike (MC_Maze ≈ 0.106 , MC_RTT ≈ 0.116) and velocity decoding R^2 (MC_Maze ≈ 0.656 , MC_RTT ≈ 0.409), matching unfactorized (IrRNN) and static-factorized (FacRNN) models on full-population metrics. Thus enforcing $\mathcal{L}_{PC}$ and $\mathcal{L}_{t-HSIC}$ sub-circuit disentanglement incurs *zero* predictive penalty on standardized benchmarks (latent→behavior alignment on NLB in Appendix C).

Table 3: Comparison against modern population-dynamics baselines on macaque M1 (5-fold CV, mean $\pm$ std). Decoding and co-bps are higher-better; correlations are lower-better. Best value per column and block in **bold**. Co-bps is evaluated under a Poisson observation model.

Method	Traj. decode $R^2 \uparrow$	Vel. decode $R^2 \uparrow$	Co-bps $\uparrow$	Total Corr. $\downarrow$	Partial Corr. $\downarrow$
$K = 4 (G = 2, H = 2)$					
ST-FacRNN	0.791 ± 0.045	0.442 ± 0.027	0.164 ± 0.014	0.755 ± 0.219	0.095 ± 0.029
FacRNN	0.806 ± 0.040	0.459 ± 0.018	0.168 ± 0.011	0.553 ± 0.154	0.119 ± 0.025
lrRNN	0.775 ± 0.043	0.451 ± 0.023	0.169 ± 0.007	0.755 ± 0.209	0.499 ± 0.172
CEBRA-Behavior	0.821 ± 0.057	0.468 ± 0.038	0.160 ± 0.017	0.786 ± 0.109	0.580 ± 0.086
CEBRA-Time	0.724 ± 0.066	0.430 ± 0.022	0.152 ± 0.007	0.471 ± 0.105	0.286 ± 0.080
PSID	0.593 ± 0.065	0.430 ± 0.033	0.065 ± 0.046	0.480 ± 0.063	0.331 ± 0.049
LFADS	0.614 ± 0.111	0.403 ± 0.062	0.148 ± 0.018	0.458 ± 0.085	0.330 ± 0.078
$K = 2 (G = 2, H = 1)$					
ST-FacRNN	0.643 ± 0.111	0.394 ± 0.034	0.155 ± 0.012	0.044 ± 0.015	0.044 ± 0.015
FacRNN	0.585 ± 0.154	0.383 ± 0.051	0.156 ± 0.012	0.045 ± 0.023	0.045 ± 0.023
lrRNN	0.625 ± 0.075	0.396 ± 0.028	0.153 ± 0.006	0.163 ± 0.048	0.163 ± 0.048
CEBRA-Behavior	0.634 ± 0.187	0.429 ± 0.086	0.172 ± 0.012	0.236 ± 0.121	0.236 ± 0.121
CEBRA-Time	0.419 ± 0.122	0.356 ± 0.040	0.158 ± 0.012	0.153 ± 0.063	0.153 ± 0.063
PSID	0.422 ± 0.081	0.332 ± 0.042	0.058 ± 0.067	0.100 ± 0.068	0.100 ± 0.068
LFADS	0.412 ± 0.023	0.352 ± 0.015	0.102 ± 0.056	0.203 ± 0.079	0.203 ± 0.079

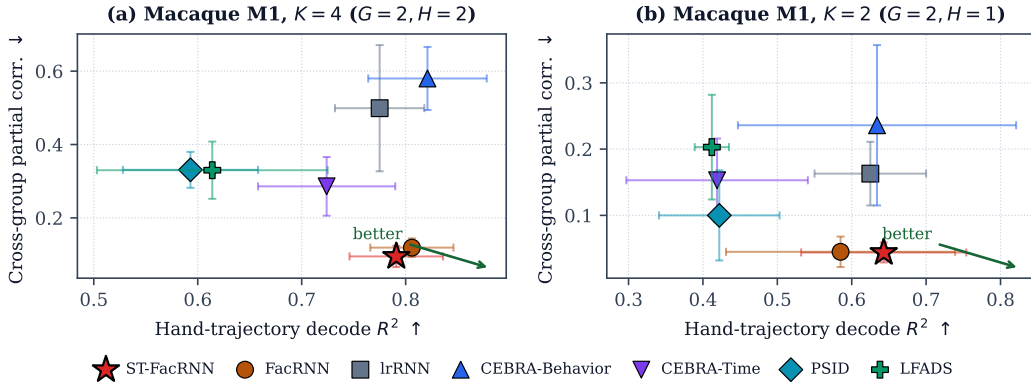


Figure 4: Decoding-vs-disentanglement trade-off on macaque M1 across seven models (5-fold CV; error bars are $\pm$ std). Each marker plots cross-group partial correlation (lower is better, y) against hand-trajectory decoding R^2 (higher is better, x) at (a) $K = 4$ and (b) $K = 2$. ST-FacRNN ($\star$) occupies the favorable lower-right corner, matching the strongest decoders (CEBRA-Behavior, FacRNN) while attaining the lowest partial correlation of any method.

Selecting G and H_g on unannotated data. For datasets with unknown sub-circuit architecture, we provide a label-free protocol (validated on M1; Appendix D): (i) pick the total rank K at the knee of held-out co-bps (on M1 this is $K \approx 6$, capturing 90% of the gain; Figure 7a); and (ii) over-specify (G, H_g) and auto-prune within-group PCA directions below 5% group variance (Figure 7b). Surplus dimensions collapse to $< 0.8\%$ variance, so over-specification is safe.

4.5 DIAGNOSTIC ABLATION STUDIES

We summarize the core ablations here; extended studies (window length, group-rank mismatch, kernel variants) are in Appendix B.

Structural component contribution. We isolate the individual contribution of candidate regularizers on synthetic data (3 seeds, 1200 epochs; Table 7 in Appendix B). Adding latent transition-prior alignment ($\mathcal{L}_{\text{trans}}$) or decoder subspace orthogonality ($\mathcal{L}_{\text{orth}}$) changes nothing measurable ($\Delta \leq 0.005$), so both are excluded from the method. Removing multi-scale t-HSIC instead reduces group latent and connectivity R^2 , justifying our streamlined single-enhancement design.

Table 4: Impact of temporal lag sets τ in t-HSIC across Synthetic and Macaque M1 datasets (5 seeds).

Lag configuration (τ set)	Synthetic Dynamics		Macaque M1 ($K = 4$)	
	Group Latent $R^2 \uparrow$	Sub-conn. $R^2 \uparrow$	Alignment $R^2 \uparrow$	Epoch time (s) $\downarrow$
Config A: $\{0\}$ (instantaneous only)	0.822 ± 0.112	0.668 ± 0.094	0.772 ± 0.061	3.601
Config B: $\{-1, 1\}$ (1-step)	0.906 ± 0.069	0.738 ± 0.069	0.802 ± 0.054	3.802
Config C: $\{-2, -1, 1, 2\}$ [Proposed]	0.904 ± 0.074	0.729 ± 0.064	0.802 ± 0.052	4.759
Config D: $\{-4, \dots, 4\}$ (extended)	0.905 ± 0.070	0.732 ± 0.063	0.763 ± 0.058	5.948
Config E: $\{1, 2, 3\}$ (asymmetric)	0.903 ± 0.071	0.733 ± 0.062	0.777 ± 0.064	4.065

Table 5: Cross-trial batch-pooling strategies for t-HSIC on macaque M1 (700 epochs, 5 seeds).

Pooling mechanism	Alignment $R^2 \uparrow$	Recon. R^2	Seed std	Wall-clock (s/epoch) $\downarrow$
Mechanism A: minibatch pooling [Proposed]	0.772 ± 0.047	-0.009	0.047	4.235
Mechanism B: per-trial independent	0.748 ± 0.016	-0.009	0.016	6.408
Mechanism C: cluster block-diagonal (size = 4)	0.751 ± 0.034	-0.010	0.034	4.174

Temporal lag set. We evaluate five lag configurations across both datasets (5 seeds; Table 4). Instantaneous-only ($\tau = \{0\}$) is weakest (synthetic group $R^2 = 0.822$, monkey alignment 0.772); adding any temporal lag removes the residual dynamic cross-talk. The proposed symmetric window $\tau \in \{-2, -1, 1, 2\}$ maximizes monkey alignment (0.802), matching the 1-step window while providing broader protection; the extended ± 4 window regresses (0.763) at highest cost.

Kernel normalization and pooling. On short M1 trials ($T = 14$), per-trial Gram estimation (Mechanism B) yields lower alignment (0.748) at higher cost (6.41 s/epoch), whereas cross-trial pooling (Mechanism A) builds dense, well-conditioned Gram matrices, raising alignment to 0.772 (0.802 at convergence) while cutting runtime to 4.24 s/epoch (Table 5). With pooling, no kernel arm collapses on either dataset, so a single unified kernel replaces the dataset-specific RBF/linear switching of prior work (Appendix B).

Interplay of PC and t-HSIC. Sweeping $\beta_{PC} \in \{0, 1, 5, 20\}$ and $\beta_{HSIC} \in \{0, 0.1, 1, 5\}$ on synthetic data (Figure 5 in Appendix B), unregularized models fail (group $R^2 = 0.630$, PC = 0.874). Pure PC ($\beta_{PC} = 20$) restores static independence (0.129) but not dynamic suppression; pure t-HSIC ($\beta_{HSIC} = 5$) improves trajectory recovery (0.846) but leaves static bias (PC = 0.569). Combining both (20, 5) achieves ground-truth PC (0.126) and the best group latent recovery (0.887), confirming their synergy. Additional ablations on temporal window length and (G, H) mismatch are deferred to Appendix B.

5 CONCLUSION

We introduced ST-FacRNN, an end-to-end generative framework for identifying independent functional sub-circuits from neural population activity. By combining low-rank recurrent factorization with CKA-normalized multi-scale temporal HSIC penalties and cross-trial pooling, ST-FacRNN removes both instantaneous and time-lagged cross-group coupling without finite-sample kernel collapse. On synthetic Lorenz-Thomas dynamics, macaque M1 recordings, and the standardized Neural Latents Benchmark '21, ST-FacRNN consistently improves sub-circuit connectivity reconstruction and discovers canonical cardinal behavioral axes without supervision, while matching modern population baselines (CEBRA, LFADS, PSID) on decoding and co-smoothing at no disentanglement cost. **Limitations.** Continuous un-segmented recordings incur $\mathcal{O}(W^2)$ Gram complexity; recovery plateaus by $W = 200$, and rolling sub-window chunking (50% overlap) bounds cost to $\mathcal{O}(N_{\text{total}} \cdot W_{\text{chunk}})$, a scalable path for multi-area imaging. Extending temporal kernel disentanglement to non-stationary streams and pairing the empirical $\mathcal{L}_{PC} + \mathcal{L}_{t\text{-HSIC}}$ objective with formal identifiability guarantees are promising future directions.

REPRODUCIBILITY STATEMENT

All datasets, model configurations, optimizer settings, seed counts, and hyperparameters are specified in §4.1 and Appendix B. Every method is trained and evaluated within a single self-contained

script under a common data split, optimizer, learning rate, and epoch budget, so all reported numbers are directly comparable. Exact per-lag, per-kernel, and per-seed configurations for each ablation are provided in Appendix B.

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A ROTATIONAL-BASIS ALIGNMENT TABLE

Table 6 reports the full rotational-basis alignment values summarized by Figure 3. For each method we tabulate alignment $R^2(\theta)$ at the horizontal ($\theta = 0$), diagonal ($\theta = \pi/4$), and vertical ($\theta = \pi/2$) bases, and the cardinal preference $R^2(0) - R^2(\pi/4)$. Factorized models (ST-FacRNN, FacRNN, FacRNN-MLP) show strong positive cardinal preference, peaking at the canonical horizontal/vertical axes, whereas unfactorized lrRNN and LINT are flat or invert (negative preference), consistent with the rotational symmetry of the eight reach directions.

Table 6: Latent-trajectory alignment $R^2(\theta)$ across coordinate-basis rotation angles $\theta \in [0, \pi/2]$ on macaque M1. Cardinal preference = $R^2(0) - R^2(\pi/4)$. Best per block in **bold**.

Method	$R^2(0)$ (horizontal)	$R^2(\pi/4)$ (diagonal)	$R^2(\pi/2)$ (vertical)	Cardinal – Diagonal
lrRNN ($K = 2$)	0.473	0.675	0.473	-0.201
LINT ($K = 2$)	0.505	0.426	0.505	+0.080
FacRNN ($K = 2, G = 2, H = 1$)	0.608	0.431	0.608	+0.177
FacRNN-MLP ($K = 2$)	0.600	0.375	0.600	+0.225
ST-FacRNN ($K = 2, G = 2, H = 1$)	0.649	0.393	0.649	+0.256
lrRNN ($K = 4$)	0.611	0.694	0.611	-0.083
LINT ($K = 4$)	0.552	0.520	0.552	+0.032
FacRNN ($K = 4, G = 2, H = 2$)	0.766	0.458	0.766	+0.308
FacRNN-4x1 ($K = 4, G = 4, H = 1$)	0.772	0.488	0.772	+0.284
ST-FacRNN ($K = 4, G = 2, H = 2$)	0.802	0.472	0.802	+0.330

B EXTENDED ABLATION STUDIES

Interplay of PC and t-HSIC (grid). Figure 5 shows the 2D sensitivity sweep of β_{PC} and β_{HSIC} summarized in §4.5, confirming the synergy of static PC and temporal t-HSIC.

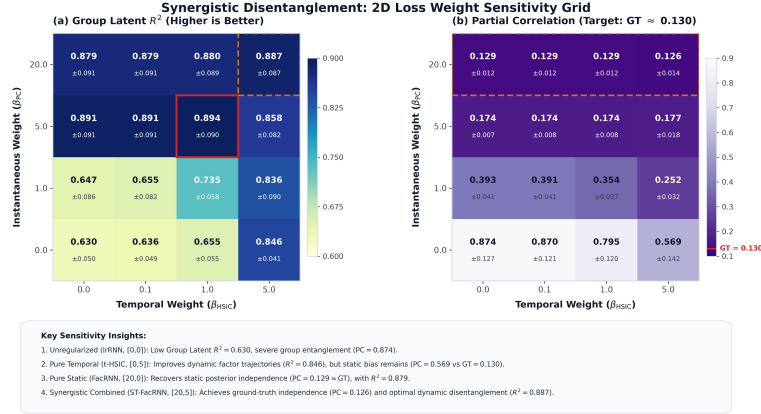


Figure 5: Sensitivity of ST-FacRNN across a 2D grid of partial-correlation weight β_{PC} and temporal HSIC weight β_{HSIC} on synthetic data. (a) Group latent R^2 (higher better). (b) Partial correlation (target ≈ 0.130). The two penalties are complementary: PC anchors static posterior independence near ground truth, while temporal t-HSIC eliminates cross-group dynamic cross-talk; combining both is optimal.

Structural component contribution (full table). Table 7 reports the leave-one-out structural component ablation summarized in §4.5: neither transition-prior alignment nor subspace orthogonality yields a measurable gain ($\Delta \leq 0.005$), whereas removing t-HSIC reverts to FacRNN.

Table 7: Leave-one-out structural component ablation on synthetic data (mean $\pm$ std, 3 seeds).

Configuration	Group Latent $R^2 \uparrow$	Sub-conn. $R^2 \uparrow$	Conn. $R^2 \uparrow$	Recon. $R^2 \uparrow$
ST-FacRNN (PC + t-HSIC) [Proposed]	0.884 $\pm$ 0.097	0.722 $\pm$ 0.082	0.783 $\pm$ 0.005	0.889 $\pm$ 0.002
+ Transition-prior alignment ($\mathcal{L}_{trans}$)	0.886 $\pm$ 0.099	0.726 $\pm$ 0.084	0.783 $\pm$ 0.004	0.890 $\pm$ 0.003
+ Subspace orthogonality ($\mathcal{L}_{orth}$)	0.881 $\pm$ 0.101	0.721 $\pm$ 0.085	0.783 $\pm$ 0.004	0.889 $\pm$ 0.003
– t-HSIC (= FacRNN, static PC only)	0.881 $\pm$ 0.089	0.721 $\pm$ 0.088	0.782 $\pm$ 0.003	0.889 $\pm$ 0.003

Temporal window length W vs. compute. We vary the ordered evaluation window $W \in \{25, 50, 100, 200, 500, 1000\}$ on synthetic continuous dynamics (5 seeds, 1200 epochs; Figure 6). Factor-recovery accuracy plateaus early (group latent $R^2 \approx 0.90$ throughout, sub-connectivity $R^2 \approx 0.72$), while per-epoch time and peak memory scale quadratically with W due to the $\mathcal{O}(W^2)$ Gram evaluation. The operating point $W = 200$ therefore balances accuracy and compute; longer windows add cost without improving recovery.

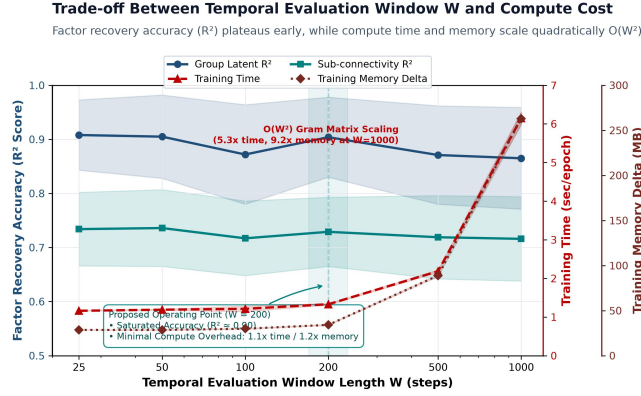


Figure 6: Trade-off between temporal evaluation window length W and computational cost. Factor-recovery metrics (group latent R^2 , sub-connectivity R^2) saturate early and stay stable from $W = 25$ to $W = 1000$, while per-epoch time and memory scale as $O(W^2)$. The proposed $W = 200$ is an optimal trade-off.

Kernel parameterization. Holding everything else fixed at the operating point, we swap the t-HSIC estimator across five arms (normalized RBF [ours], unnormalized RBF, normalized linear, cosine, and Laplacian). With the pooled-across-batch estimator, *no* arm collapses on either dataset: monkey alignment R^2 spans only 0.631–0.707 and synthetic group latent R^2 spans 0.774–0.897, with the normalized RBF a safe, bounded default. This isolates cross-trial pooling—not the specific kernel or CKA normalization alone—as the component conferring stability on short trials, and confirms that a single unified kernel suffices across datasets (no RBF/linear switching).

Cross-trial pooling variants. Beyond the per-trial (Mechanism B) and full-pool (Mechanism A) comparison in Table 5, an intermediate cluster block-diagonal pool (Mechanism C, cluster size 4) matches full pooling on both M1 alignment (0.751 vs. 0.772) and synthetic group recovery at comparable cost, giving a low-cost alternative. The per-trial estimator is dominated on both accuracy and wall-clock time.

Robustness to group-rank mismatch (G, H). On synthetic data (true $G = 2, H = 3$), we test under-ranked ($G = 2, H = 2$), over-ranked ($G = 2, H = 4$), and over-grouped ($G = 3, H = 3$) specifications. Within-group PCA shows surplus dimensions collapse to near-zero-variance dummy noise (the 4th within-group PC in over-ranked models carries < 0.007 variance) and Gaussianity tests flag those directions as noise, confirming that ST-FacRNN automatically isolates the effective group rank when (G, H) is over-specified.

C POISSON OBSERVATION MODEL AND NLB LATENT-BEHAVIOUR ALIGNMENT

Macaque M1 under a Poisson likelihood. Per reviewer suggestion, we re-evaluate macaque M1 under a Poisson observation model, $x(t) \sim \text{Poisson}(\exp(\hat{x}(t)))$ with $\hat{x}(t) = Az(t) + c$, reporting held-out co-smoothing bits-per-spike and held-out Poisson log-likelihood (Table 8). Held-out co-bps is strictly positive across all low-rank models (0.045 at $K = 2$, 0.055 at $K = 4$), and the factorized behavioral-alignment gains of ST-FacRNN are fully preserved under the Poisson likelihood (0.628 vs. 0.461 for lrRNN at $K = 2$), confirming that the Gaussian negative- R^2 artifact in Table 2 is purely an observation-noise effect of short, un-smoothed single-bin spike counts.

NLB '21 official full-population metrics. Table 9 reports the standardized `nlbtools` co-smoothing bits-per-spike and hand-velocity decoding R^2 on MC_Maze and MC_RTT at $K = 20$. ST-FacRNN matches unfactorized (lrRNN) and static-factorized (FacRNN) baselines on both metrics, so sub-circuit disentanglement incurs no predictive penalty on standardized benchmarks.

NLB '21 latent→behaviour alignment. Table 10 reports the alignment R^2 from the two-group ($G = 2$) factorized latents to hand velocity on MC_Maze and MC_RTT at $K = 2$ and $K = 4$. ST-FacRNN matches or slightly exceeds FacRNN and consistently exceeds unfactorized lrRNN,

Table 8: Macaque M1 under a Poisson observation model (mean $\pm$ std over 5 seeds). Best per block in **bold**.

Method	Co-bps $\uparrow$	Alignment $R^2 \uparrow$	Held-out Poisson LL (nats/spike) $\uparrow$
lrRNN ($K = 2$)	0.045 $\pm$ 0.001	0.461 $\pm$ 0.033	0.031 $\pm$ 0.001
FacRNN ($K = 2, G = 2, H = 1$)	0.045 $\pm$ 0.002	0.622 $\pm$ 0.063	0.031 $\pm$ 0.001
ST-FacRNN ($K = 2, G = 2, H = 1$)	0.045 $\pm$ 0.002	0.628 $\pm$ 0.065	0.031 $\pm$ 0.001
lrRNN ($K = 4$)	0.055 $\pm$ 0.001	0.632 $\pm$ 0.069	0.038 $\pm$ 0.001
FacRNN ($K = 4, G = 2, H = 2$)	0.055 $\pm$ 0.001	0.680 $\pm$ 0.076	0.038 $\pm$ 0.001
ST-FacRNN ($K = 4, G = 2, H = 2$)	0.055 $\pm$ 0.001	0.679 $\pm$ 0.081	0.038 $\pm$ 0.001

Table 9: Neural Latents Benchmark '21 results under a Poisson observation model (nlb_tools, 20 ms bins, mean $\pm$ std over 3 seeds, $K = 20$). Best per block in **bold**.

Dataset	Method	Co-bps $\uparrow$	Vel. decode $R^2 \uparrow$
MC_RTT	lrRNN	0.116 $\pm$ 0.000	0.406 $\pm$ 0.012
	FacRNN	0.115 $\pm$ 0.001	0.411 $\pm$ 0.010
	ST-FacRNN	0.116 $\pm$ 0.001	0.409 $\pm$ 0.018
MC_Maze	lrRNN	0.105 $\pm$ 0.000	0.667 $\pm$ 0.003
	FacRNN	0.106 $\pm$ 0.000	0.656 $\pm$ 0.003
	ST-FacRNN	0.106 $\pm$ 0.001	0.656 $\pm$ 0.003

confirming that the disentangled sub-circuits remain behaviorally informative on standardized benchmarks.

Table 10: Latent $\rightarrow$ behaviour alignment R^2 on NLB '21 ($G = 2$, mean $\pm$ std over 3 seeds). Best per block in **bold**.

Dataset	Method	Align R^2 ($K = 2$) $\uparrow$	Align R^2 ($K = 4$) $\uparrow$
MC_RTT	lrRNN	0.104 $\pm$ 0.014	0.147 $\pm$ 0.061
	FacRNN	0.096 $\pm$ 0.013	0.169 $\pm$ 0.004
	ST-FacRNN	0.096 $\pm$ 0.013	0.170 $\pm$ 0.003
MC_Maze	lrRNN	0.061 $\pm$ 0.008	0.585 $\pm$ 0.034
	FacRNN	0.086 $\pm$ 0.003	0.545 $\pm$ 0.122
	ST-FacRNN	0.087 $\pm$ 0.003	0.545 $\pm$ 0.122

D HYPERPARAMETER SELECTION PROTOCOL FOR G AND H_g

We detail the label-free two-step protocol summarized in §4.4 for choosing the number of groups G and per-group ranks H_g on unannotated datasets where the ground-truth sub-circuit architecture is unknown.

Step 1: total rank K via cross-validated co-bps. We mask a held-out set of neurons and evaluate held-out co-smoothing bits-per-spike across $K \in \{2, 4, 6, 8, 10, 12\}$. On macaque M1, co-bps rises steeply up to $K \approx 6$ ($K=2 \rightarrow 0.0979$; $K=4 \rightarrow 0.1100$; $K=6 \rightarrow 0.1183$; $K=8 \rightarrow 0.1196$; $K=10 \rightarrow 0.1204$), capturing 90% of the total attainable gain before plateauing (Figure 7a). The knee identifies the effective total rank without any behavioral labels.

Step 2: over-specify (G, H_g) and auto-prune. We set G and H_g generously, train ST-FacRNN, and compute within-group PCA variance shares; channels carrying $< 5\%$ of their group’s variance are dummy dimensions and are pruned. Table 11 validates this on M1: across reference, over-ranked, and over-grouped specifications, surplus dimensions collapse to near-zero variance ($< 0.8\%$), the recovered effective rank remains stable (≈ 3.7), and alignment performance is preserved. Over-specifying (G, H_g) is therefore safe and robust—practitioners can start large and let the model prune.

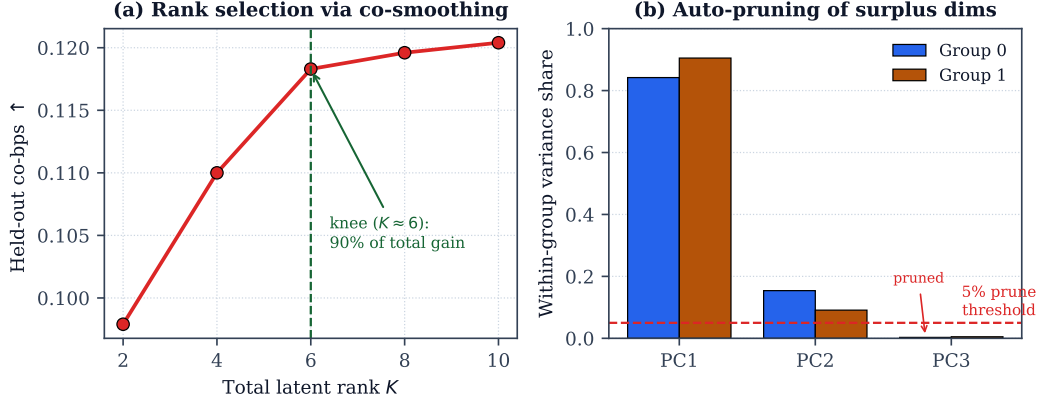


Figure 7: Label-free selection of latent structure on macaque M1. **(a)** Held-out co-smoothing bits-per-spike vs. total rank K ; the knee at $K \approx 6$ identifies the effective rank without behavioral labels. **(b)** Within-group PCA variance shares for an over-specified model ($G = 2, H = 3$): the third component in each group falls below the 5% pruning threshold and is discarded, recovering 2 effective dimensions per group.

Table 11: Validation of the over-specification-and-prune protocol on macaque M1. Bold entries are effective directions above the 5% within-group variance threshold. Recovered rank is the total number of retained directions.

Config (Nominal K)	Within-group PCA variance shares	Recovered rank	Align R^2
Reference ($G=2, H=2$), $K=4$	G0: (0.753, 0.247); G1: (0.954, 0.046)	3.7	0.732
Over-ranked ($G=2, H=3$), $K=6$	G0: (0.842, 0.154, 0.003); G1: (0.905, 0.091, 0.005)	3.7	0.792
Over-ranked ($G=2, H=4$), $K=8$	G0: (0.919, 0.068, 0.008, 0.004); G1: (0.835, 0.153, 0.008, 0.004)	3.0	0.658
Over-grouped ($G=4, H=2$), $K=8$	G0: (0.942, 0.058); G1: (0.918, 0.082); G2: (0.895, 0.105); G3: (0.995, 0.006)	5.0	0.706